# 🛰️ MAITRI

Multimodal AI Assistant for Psychological & Physical Well-Being of Astronauts

SIH 2025 | HARD | Problem ID: 25175 | ISRO / Dept. of Space

## 1 Problem Summary

Astronauts aboard the **Bhartiya Antariksh Station (BAS)** face isolation, sleep disruption, tight schedules, and physical discomforts — all of which can trigger serious psychological and physical issues. **MAITRI** is a **standalone, offline, multimodal AI assistant** that:

1. **Detects** emotions via facial expressions + voice tone (audio-video)
2. **Provides** short adaptive conversations, psychological companionship & evidence-based interventions
3. **Reports** critical emotional/physical states to ground control

> ⚠️ **KEY CONSTRAINT**
>
> The deliverable must be a **trained AI model running on a standalone offline system** (edge computing). No cloud dependency in production.

## 2 High-Level Architecture

### End-to-End System Architecture

🎥 Camera Feed
Video Stream

🎤 Microphone Feed
Audio Stream

▼

Face Detection & Tracking

Voice Activity Detection

Speech-to-Text (Whisper)


Facial Emotion Recognition
Speech Emotion Recognition
Text Sentiment Analysis
Multimodal Fusion Module
Attention-weighted late fusion
Emotion State Tracker
Risk Scoring & Thresholds
Context Memory (Session History)
Local LLM (Conversational AI)
Intervention Selector
Text-to-Speech (Offline TTS)
Audio Response
Local Dashboard
Ground Station Alerts
Session Logs

## 3 Module-by-Module Breakdown

### 3.1 Input Layer

| COMPONENT | PURPOSE | TECHNOLOGY |
|---|---|---|
| **Camera Feed** | Continuous video capture of astronaut's face | OpenCV VideoCapture, USB/IP Camera |
| **Microphone Feed** | Continuous audio capture of astronaut's voice | PyAudio / sounddevice |

📝 **NOTE**

Feeds are processed in real-time with frame-level and chunk-level buffering. Video at ~15 FPS (sufficient for FER), audio at 16kHz mono.

### 3.2 Preprocessing Pipeline

| COMPONENT | WHAT IT DOES | MODEL / LIBRARY |
|---|---|---|
| **Face Detection & Tracking** | Locates and tracks face across frames | MediaPipe Face Mesh / MTCNN / RetinaFace |
| **Facial Landmark Extraction** | Extracts 468 landmarks for AU detection | MediaPipe / dlib 68-point |
| **Voice Activity Detection** | Filters silence, isolates speech segments | Silero VAD (lightweight, offline) |
| **Audio Feature Extraction** | Extracts MFCCs, Mel-spectrograms, pitch, energy | Librosa / torchaudio |
| **Speech-to-Text** | Transcribes speech for NLP analysis | OpenAI Whisper (tiny/base — offline) |

### 3.3 Multimodal Analysis Engine (Core AI)

This is the heart of MAITRI — three parallel emotion recognition streams fused into a unified emotional state.

#### 3.3.1 Facial Emotion Recognition (FER)

```
Input:   Cropped face frames (48×48 or 224×224)
Model:   Fine-tuned MobileNetV2 / EfficientNet-Lite / Custom CNN
Output:  Probabilities over 7 emotions
         [Happy, Sad, Angry, Fear, Surprise, Disgust, Neutral]
Dataset: FER2013, AffectNet, RAF-DB
```

### 3.3.2 Speech Emotion Recognition (SER)

```
Input:   Mel-spectrograms / MFCC features from audio chunks (2-5s)
Model:   CNN-LSTM hybrid / HuBERT-tiny fine-tuned
Output:  Probabilities over emotions
         [Happy, Sad, Angry, Fear, Neutral, Stressed, Fatigued]
Dataset: RAVDESS, CREMA-D, IEMOCAP, EMO-DB
```

### 3.3.3 Text Sentiment Analysis

```
Input:   Transcribed text from Whisper STT
Model:   DistilBERT / TinyBERT fine-tuned for sentiment + emotion
Output:  Sentiment (Positive/Negative/Neutral) + Emotion labels
Dataset: GoEmotions, EmoInt, custom astronaut dialogue corpus
```

### 3.3.4 Multimodal Fusion Module

```
Strategy: Late Fusion with Attention-weighted aggregation

Fusion Formula:
  E_final = α·E_face + β·E_voice + γ·E_text

  Where α, β, γ are learned attention weights
  (or confidence-weighted based on input quality)

Conflict Resolution:
  - If face says "happy" but voice says "stressed" →
    voice gets higher weight (harder to mask)
  - Missing modality graceful degradation
    (e.g., face not visible → rely on audio+text)
```

### 3.3.5 Physical Distress Detector

| SIGNAL | DETECTION METHOD |
|---|---|
| **Fatigue** | Yawning detection (mouth aspect ratio), eye blink rate, droopy eyelids |
| **Pain / Discomfort** | Facial Action Units (AU4, AU6, AU7, AU9, AU10, AU43) |
| **Sleep Deprivation** | Microsleep detection (prolonged eye closure), response latency |

| SIGNAL | DETECTION METHOD |
| --- | --- |
| **Stress** | Voice jitter, pitch variability, speech rate changes |

## 3.4 Decision & State Engine

### 3.4.1 Emotion State Tracker

Maintains a **rolling temporal profile** of the astronaut's emotional state:

```
EmotionState:
  current_emotion:   "stressed"
  confidence:        0.82
  duration_in_state: "00:14:32"
  trend:             "worsening"   # stable / improving / worsening
  history:           [last 24h rolling window]
  baseline:          {personal average emotional profile}
```

### 3.4.2 Risk Scoring & Escalation Levels

🟢 **Level 0: Normal (Score 0–30)**

Astronaut shows stable, healthy emotional baseline

→ Passive monitoring, log only

🟡 **Level 1: Mild Concern (Score 31–50)**

Slight emotional deviation detected

→ Proactive check-in, gentle conversation

🟠 **Level 2: Moderate Risk (Score 51–70)**

Sustained negative state or physical distress

→ Active intervention + Alert ground station

🔴 **Level 3: Critical (Score 71–100)**

Severe emotional distress or crisis indicators

→ Emergency: Immediate ground notification + Crisis protocol

**Risk Score Formula:**

```
risk_score = w1 × emotion_severity
           + w2 × duration_factor
           + w3 × trend_factor
           + w4 × physical_distress_score
           + w5 × context_modifier (time of day, mission phase, etc.)
```

### 3.4.3 Context Memory

- **Short-term:** Current conversation session (what was discussed, astronaut's responses)
- **Long-term:** SQLite database with historical emotion logs, intervention history, personal preferences
- **Astronaut Profile:** Baseline mood patterns, known triggers, preferred coping strategies

## 3.5 Response Engine

### 3.5.1 Conversational AI (Local LLM Options)

| OPTION | MODEL SIZE | SUITABILITY |
|---|---|---|
| **Phi-3 Mini (3.8B)** | ~2.5 GB quantized | ⭐ Best balance of quality + edge performance |
| **Gemma 2B** | ~1.5 GB quantized | Lighter alternative |
| **Llama 3.2 3B** | ~2 GB quantized | Strong general capability |
| **TinyLlama 1.1B** | ~700 MB quantized | Ultra-lightweight fallback |

**System Prompt Strategy:**

```
You are MAITRI, a compassionate AI companion for astronauts aboard
the Bhartiya Antariksh Station. Your role is to:
- Provide warm, empathetic psychological support
- Keep conversations short and situation-relevant
- Use evidence-based therapeutic techniques (CBT, mindfulness, etc.)
- Never diagnose — support and escalate when needed
- Be culturally sensitive and adapt to the astronaut's preferences

Current astronaut emotional state: {emotion_state}
Risk level: {risk_level}
Suggested intervention: {intervention_type}
Conversation history: {recent_context}
```

### 3.5.2 Evidence-Based Intervention Selector

| EMOTION STATE | INTERVENTION TYPE | EXAMPLE RESPONSE |
|---|---|---|
| Anxious / Stressed | Breathing exercises, grounding | "Let's try box breathing — inhale for 4..." |
| Sad / Lonely | Empathetic listening, reminiscence | "Would you like to talk about what's on your mind?" |
| Angry / Frustrated | De-escalation, cognitive reframing | "That sounds frustrating. Let's break it down..." |
| Fatigued | Sleep hygiene, micro-rest guidance | "A 10-min power nap could help. Shall I set a timer?" |
| Normal (proactive) | Casual check-in, encouragement | "How's the experiment going? You're doing great work." |

> 💡 **TIP**
>
> Interventions are stored in a structured JSON knowledge base with tags, so the system can retrieve contextually relevant responses without internet.

### 3.5.3 Text-to-Speech (Offline)

| OPTION | NOTES |
| --- | --- |
| **Piper TTS** | Fast, offline, multiple voices, ONNX-based |
| **Coqui TTS** | Higher quality, slightly heavier |
| **eSpeak-NG** | Ultra-lightweight fallback |

## 4 Edge Deployment Architecture

### Deployment View — Standalone Edge System

**🖥️ Edge Device (Standalone)**

**Hardware:**
- NVIDIA Jetson Orin / RTX GPU
- x86/ARM CPU
- SSD Storage (256GB+)
- Camera + Microphone

**Software:**
- Ubuntu 22.04 LTS
- Python 3.10 + ONNX Runtime
- llama.cpp (LLM inference)
- FastAPI Backend
- HTML/JS Dashboard
- SQLite Database

**🌍 Ground Station**

- Alert Receiver
- Mission Control Dashboard
- Post-Mission Analytics

*Connected via queued alerts (when comm window available)*

### Hardware Requirements

| COMPONENT | MINIMUM | RECOMMENDED |
|---|---|---|
| **GPU** | NVIDIA Jetson Orin Nano (8GB) | NVIDIA Jetson AGX Orin (32GB) |
| **CPU** | ARM Cortex-A78AE / Intel i5 | 12-core ARM / Intel i7 |
| **RAM** | 16 GB | 32 GB |
| **Storage** | 128 GB SSD | 256 GB SSD |
| **Camera** | 720p USB | 1080p with IR (low-light) |
| **Microphone** | USB microphone | Noise-cancelling array mic |

## 5 Data Processing Flow

### Sequence: From Astronaut Interaction to Response

| STEP | COMPONENT | ACTION |
| --- | --- | --- |
| 1 | 📷 Camera + 🎤 Microphone | Capture video frames (15 FPS) + audio chunks (2-5s) in parallel |
| 2 | ⚙️ Preprocessor | Face detection, landmark extraction, VAD, MFCC extraction, STT transcription |
| 3 | 🧠 Analysis Engine | Run FER, SER, and Sentiment analysis in parallel |
| 4 | 🔀 Fusion Module | Attention-weighted fusion of all three modalities |
| 5 | 📊 Decision Engine | Update emotion state tracker, compute risk score |
| 6 | ⚡ Escalation Check | If Risk Level ≥ 2 → Queue ground station alert |
| 7 | 💬 Response Engine | Generate contextual response via LLM + intervention selector |
| 8 | 🔊 TTS Output | Convert response to speech and deliver to astronaut |

## 6 Project Structure

```
MAITRI/ ├── data/ │ ├── datasets/ # Training datasets (FER2013, RAVDESS, etc.) │
├── interventions/ # JSON knowledge base of interventions │ └── profiles/ #
Astronaut baseline profiles │ ├── models/ │ ├── fer/ # Facial Emotion
Recognition model │ ├── ser/ # Speech Emotion Recognition model │ ├── sentiment/
# Text sentiment model │ ├── whisper/ # Whisper STT model weights │ ├── llm/ #
Local LLM weights (quantized) │ └── tts/ # TTS model weights │ ├── src/ │ ├──
input/ │ │ ├── camera.py # Video capture & frame buffering │ │ └── microphone.py
# Audio capture & chunking │ │ │ ├── preprocessing/ │ │ ├── face_detector.py #
Face detection & landmark extraction │ │ ├── audio_processor.py # VAD, MFCC,
spectrogram extraction │ │ └── transcriber.py # Whisper STT wrapper │ │ │ ├──
analysis/ │ │ ├── fer_model.py # Facial emotion recognition │ │ ├── ser_model.py
# Speech emotion recognition │ │ ├── sentiment.py # Text sentiment analysis │ │
├── fusion.py # Multimodal fusion module │ │ └── physical.py # Physical distress
detection │ │ │ ├── decision/ │ │ ├── state_tracker.py # Emotion state
management │ │ ├── risk_scorer.py # Risk level computation │ │ └──
context_memory.py # Session & long-term memory │ │ │ ├── response/ │ │ ├──
conversation.py # LLM conversation manager │ │ ├── interventions.py # Evidence-
based intervention selector │ │ └── tts_engine.py # Text-to-speech wrapper │ │ │
├── output/ │ │ ├── dashboard.py # Local web dashboard (FastAPI) │ │ ├──
ground_alert.py # Ground station alert queue │ │ └── logger.py # Session logging
& analytics │ │ │ ├── pipeline.py # Main orchestration pipeline │ └── config.py
# Configuration & constants │ ├── frontend/ │ ├── index.html # Dashboard UI │
├── styles.css # Dashboard styling │ └── app.js # Dashboard logic (charts, real-
time) │ ├── tests/ ├── scripts/ ├── requirements.txt ├── Dockerfile └── README.md
```

## 7 Tech Stack Summary

| LAYER | TECHNOLOGY | JUSTIFICATION |
|---|---|---|
| **Language** | Python 3.10+ | Best ML/AI ecosystem |
| **Video Processing** | OpenCV, MediaPipe | Lightweight, real-time |
| **Audio Processing** | Librosa, torchaudio, Silero VAD | Industry standard, offline |
| **STT** | Whisper (tiny/base) | State-of-the-art, runs offline |
| **FER Model** | MobileNetV2 / EfficientNet-Lite | Lightweight for edge |
| **SER Model** | CNN-LSTM / HuBERT-tiny | Proven on emotion datasets |

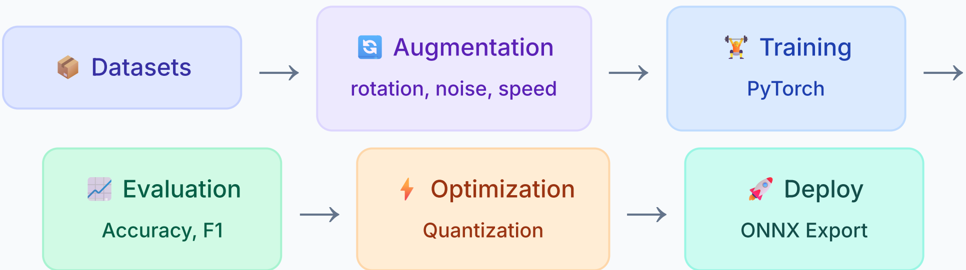
| LAYER | TECHNOLOGY | JUSTIFICATION |
|---|---|---|
| **Sentiment** | DistilBERT / TinyBERT | Small footprint, fast |
| **Fusion** | Custom attention-weighted | Handles missing modalities |
| **Local LLM** | Phi-3 Mini (3.8B, Q4) | Best quality-to-size for edge |
| **LLM Runtime** | llama.cpp / ONNX Runtime | Optimized CPU+GPU inference |
| **TTS** | Piper TTS | Fast, offline, natural |
| **Backend** | FastAPI | Async, WebSocket support |
| **Frontend** | HTML/CSS/JS + Chart.js | No build step needed |
| **Database** | SQLite | Zero-config, embedded |
| **Model Format** | ONNX / GGUF (LLM) | Cross-platform, optimized |

## 8 Training Strategy

### 8.1 Datasets

| TASK | DATASET | SIZE | NOTES |
|---|---|---|---|
| FER | FER2013 | 35K images | Baseline |
| FER | AffectNet | 450K images | Larger, higher quality |
| FER | RAF-DB | 30K images | Real-world, diverse |
| SER | RAVDESS | 7,356 clips | Clean, acted |
| SER | CREMA-D | 7,442 clips | Diverse demographics |
| SER | IEMOCAP | 12 hours | Conversational, natural |
| Sentiment | GoEmotions | 58K texts | 27 emotion categories |
| Physical | YawDD | 322 videos | Yawning / drowsiness |

### 8.2 Training Pipeline




## 9 Interaction Modes

**👁️ Passive Monitoring (Always On)**

Camera + mic continuously capture. Emotion analysis runs in background. Dashboard updates in real-time. No astronaut interaction required.

**💬 Proactive Check-in (System Initiated)**

Triggered when risk score crosses threshold or at scheduled intervals. MAITRI initiates: "Hey, I noticed you seem a bit tense..."

🗣️ **Reactive Support (Astronaut Initiated)**

Astronaut says "MAITRI" (wake word) or presses button. Full conversational mode activated.

🚨 **Crisis Mode (Emergency)**

Triggered at Risk Level 3. Immediate ground alert, crisis de-escalation protocol, continuous monitoring until state improves.

## 10 Ground Station Alert Format


```
{ "alert_id": "MAITRI-2025-08-28-001", "timestamp": "2025-08-28T14:32:00Z",
"astronaut_id": "CREW-01", "risk_level": 3, "risk_score": 78,
"emotional_state": { "primary": "distressed", "confidence": 0.85,
"duration_minutes": 45, "trend": "worsening" }, "physical_state": {
"fatigue_level": "high", "sleep_quality_24h": "poor", "pain_indicators": false
}, "intervention_attempted": { "type": "breathing_exercise", "response":
"partially_engaged", "effectiveness": 0.3 }, "recommended_action": "Schedule
counseling session with ground psychologist" }
```


## 11 Key Differentiators

| FEATURE | WHY IT MATTERS |
|---|---|
| **Multimodal Fusion with graceful degradation** | Works even if camera is blocked or mic is noisy |
| **Temporal emotion tracking** | Detects prolonged states, not just snapshots |
| **Evidence-based interventions** | Uses CBT, mindfulness, grounding — not just chatting |
| **Fully offline** | Zero latency, works in space with no connectivity |
| **Astronaut personalization** | Learns individual baselines and preferences over time |
| **Physical + Psychological** | Detects fatigue, pain, sleep issues — not just emotions |
| **4-tier risk escalation** | Proportional response from passive to emergency |

| FEATURE | WHY IT MATTERS |
| --- | --- |
| **Ground station integration** | Structured alerts for mission control psychologists |

## 12 Development Phases (Hackathon Timeline)

### Phase 1: Foundation — Day 1, First Half

- ☐ Set up project structure & environment
- ☐ Implement camera + microphone input capture
- ☐ Face detection + landmark extraction pipeline
- ☐ Audio preprocessing (VAD, MFCC extraction)

### Phase 2: Core AI Models — Day 1, Second Half

- ☐ Train / fine-tune FER model on FER2013/AffectNet
- ☐ Train / fine-tune SER model on RAVDESS/CREMA-D
- ☐ Integrate Whisper for STT
- ☐ Implement text sentiment analysis

### Phase 3: Fusion & Decision — Day 2, First Half

- ☐ Build multimodal fusion module
- ☐ Implement emotion state tracker
- ☐ Build risk scoring system
- ☐ Physical distress detection (fatigue, yawning)

### Phase 4: Response & Conversation — Day 2, Second Half

- ☐ Set up local LLM (Phi-3 Mini quantized)
- ☐ Build intervention knowledge base
- ☐ Implement conversation manager with context
- ☐ Integrate Piper TTS for spoken responses

### Phase 5: Integration & UI — Day 3, First Half

- ☐ Build real-time dashboard (emotion trends, risk levels)
- ☐ End-to-end pipeline integration
- ☐ Ground station alert system
- ☐ Session logging

### Phase 6: Polish & Demo — Day 3, Second Half

- ☐ Testing & bug fixes
- ☐ Performance optimization for edge
- ☐ Demo preparation & presentation
- ☐ Documentation

## 13 Evaluation Metrics

| METRIC | TARGET | MEASUREMENT |
|---|---|---|
| **FER Accuracy** | ≥ 70% on 7-class | Test set accuracy + F1 |
| **SER Accuracy** | ≥ 65% on 6-class | Test set accuracy + F1 |
| **Fusion Accuracy** | ≥ 75% combined | Cross-modal validation |
| **Response Latency** | < 3 seconds E2E | Time from detection to speech |
| **False Alert Rate** | < 10% | Incorrect escalations |
| **Conversation Quality** | Subjective eval | Empathy, relevance, helpfulness |
| **System Uptime** | 99%+ continuous | Stress test over 24 hours |